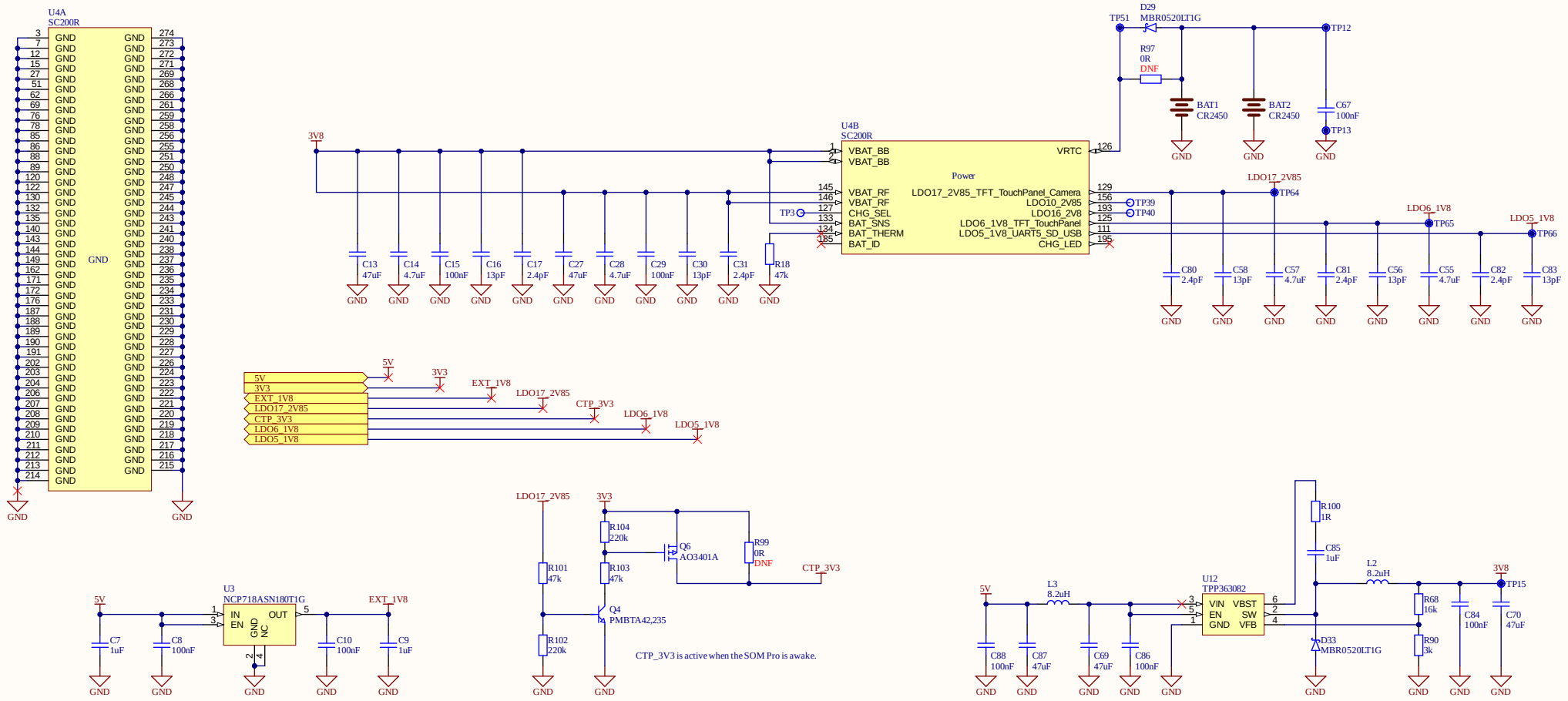


Power

SC200R-WF
The SC200R requires a 3A, 3.8V nominal supply.
Max WiFi power is 1W as opposed to mobile phone power, that is 2W, consequently the SC200R-WF probably only requires a 1.5A, 3.8V nominal supply.
In off mode: 8uA typical.
In Airplane mode: 3.2mA typical, 2.6mA measured.
During Bluetooth data transfer BR (DH5) @ max powe: 92mA typical.
During WiFi data transfer, 2.4GHz, 11b 1Mbps @ max power: 143mA typical.
During WiFi data transfer, 5GHz, 11n HT40 MCS7 @ max power: 154mA typical.
Active mode, no WiFi, measured: 11mA.
Sleep mode with WiFi, measured: 10mA.
Active mode with WiFi measured: 14mA.
The module powers off if the input voltage goes below 3.1V.
SC200RWFxx-F51-xxxxx is the 8GB plus 1GB WiFi variant.
SC200RWFxx-F53-xxxxx is the 16GB plus 2GB WiFi variant.

SC200R-WF RTC
VRTC should be 2V to 3.25V.
VRTC consumes 6uA average measured 25C, 300uA peak.
Every 10.97s it consumes 310uA for 145us measured.
VRTC consumes 6.7uA average measured 50C.
VRTC charge can be turned on with "qcom,charge-enable = <1>;" property in dtsi.
VRTC charge voltage can be set to 3.1V with "qcom,vset-millivolts = <3100>;" property in dtsi. Alternatively 2.5V, 3.0V or 3.2V.
VRTC charge resistance can be set to 800ohm with "qcom,rset-ohms = <800>;" property in dtsi. Alternatively 1200ohm, 1700ohm or 2100ohm. Default 2100ohm.

SC200R-WF Power Rails
If charging is enabled, and USB is plugged in, 4.2V can appear on the VBAT_BB pins.
VBAT_BB and VBAT_RF should be 3.55V to 4.2V, 3.8V nominal. 3A total max in.
LDO5_1V8 remains on in sleep mode. Produces 20mA max.
LDO5_1V8 wakes up 47ms after a falling edge on the PWRKEY pin.
PWRKEY is internally pulled up to 1.8V (we don't know which 1.8V).
The module is woken up by holding the PWRKEY low (through 1k) for at least 1.6s.
Should not attempt to drive the PWRKEY low for less than 30ms after the VBAT pins are stable.
Should not hold PWRKEY low indefinitely.
LDO6_1V8 produces 150mA max.
LDO6_1V8 and LDO17_2V85 wakes up periodically in sleep mode under software control.
LDO6_1V8 and LDO17_2V85 goes back to sleep under software control.
LDO6_1V8 and LDO17_2V85 can wake up from a falling edge on PWRKEY with a programmable delay.
The module has an internal 1V8 LDO that comes up directly off the VBAT pins (or it could be off VBAT_BB only), that we don't have direct access to.

TPP363082
5.6A max high side current limit.
400kHz operation.
Synchronous, current mode, forced PWM controller.
Line regulation at 5V in is 1%.
Load regulation at 0A is 0.3%.
Load regulation at 1A is 0%.
Load regulation at 3A, 24V in is 0.6%.
Load regulation at 3A, 12V in is 1%.
Efficiency 3A, 12V in is 92%.
Loss at no load, 5V supply: 10mW.
Load transient, 0A to 3A, 5V out, 12V in, 10uF output capacitance, 10uH inductor, for the pulse skipping variant is 630mV.
Load transient, 3A to 0A, 5V out, 12V in, 10uF output capacitance, 10uH inductor, is 440mV.
With a 47uF output capacitor, the output should dip not more than 100mV on a 0 to 3A transient.

Starting the TPP363082
The TPP363081 draws 2uA typical when off.
It has an undervoltage lockout on the input voltage set to 4.4V max.
The TPP363081 depends on a minimum amount of leakage through the load to start up.
To get 4.4V to start, when the input is as low as 4.75V leaves only 0.35V accross the load.
With a 16k and 3k voltage divider at the output, 350mV produces 18.4uA of leakage into the load.
The high temperature startup capability of the circuit should be tested if it is intended to be used off a nominal 5V input.
There is no risk of the chip shutting down during low power mode because this IC does not have low power mode.
A potential alternative to the TPP363082 is the LMR51450FNDRRR.
100nF is requested as the bootstrap capacitor, but startup takes 2.5ms, in that time can only drop 0.2V, and draws 90uA, so size has been increased.